Physikalisches Anfängerpraktikum 1 (PAP1)

V11 - Introduction

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1 Introduction

1.1 Goal of the experiment

The goal of the introduction experiment was to learn basic techniques of experimenting, systematical data measuring and statistical analysis of those data. In detail the key aspects were:

- Comparison of different measuringg methods to determine the period of oscillation of the spring pendulum (start/stop at mass passing the maximum elongation vs. at the zero passing) as well as determining the uncertainties of each method.
- Graphical calculation of the spring constant D from the period of oscillation T of the mass m.
- Determining the earths accaleration g from the displacement x of the mass m.

1.2 Setup and methods of measuring

In figure 1 one can see the setup used in the experiment.

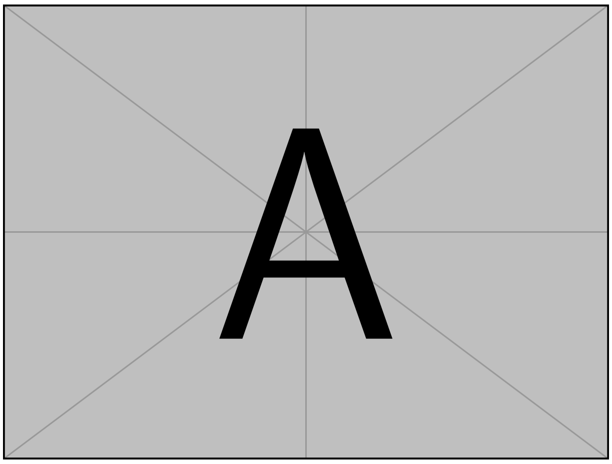

Figure 1: Unfortunatly I am missing the picture of the setup (next time I will take a picture)

1.3 Theoretical background

The equation of motion of the ideal, undamped spring pendulum is:

$$m\ddot{x} = -Dx \tag{1}$$

with m being tha attached mass, x the displacement from the initial position and D the spring constant. The solution of the ODE for the initial condition $x(0) = x_0$ is given by a cosine function:

$$x(t) = x_0 \cos(\omega t) \quad \text{mit} \quad \omega = \sqrt{\frac{D}{m}} \tag{2}$$

From the relation $\omega = \frac{2\pi}{T}$ one gets the oscillation period T:

$$T = 2\pi\sqrt{\frac{m}{D}} \tag{3}$$

Squaring equation 3 yields a linear relation of the form $y = a \cdot x + b$:

$$T^2 = \frac{4\pi^2}{D} \cdot m \tag{4}$$

The slope $a = \frac{\Delta T^2}{\Delta m}$ in the T^2-m-diagramm is therefore $a = \frac{4\pi^2}{D}$ from which one can determine the spring constant:

$$D = \frac{4\pi^2}{a} \tag{5}$$

For the displacement x of the mass m due to gravitational accelaration hook's law reads:

$$mg = Dx \iff x = \frac{g}{D} \cdot m \tag{6}$$

From the slope $a' = \frac{\Delta x}{\Delta m} = \frac{g}{D}$ in the x-m-diagramm one can then finally determine the gravitational accelaration with $g = D \cdot a'$.

2 Measurement protocol

The original signed protocol is attached below.

V Measurement protocol

Comparison of methods for determining the period of oscillation

No.	number of oscillations n	measured time t [s]	period of oscillation T [s]	mean value $\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]
1	3	4,63	1,54		
2	3	4,69	1,56		
3	3	4,81	1,60		
4	3	4,83	1,61		
5	3	4,76	1,59	1,587	$2{,}7\cdot10^{-3}$
6	3	4,77	1,59		
7	3	4,77	1,59		
8	3	4,88	1,63		
9	3	4,70	1,57		
10	3	4,76	1,59		

Start and stop at the maximum deflection

No.	number ob oscillations n	measured time t [s]	period of oscillation T [s]	mean value $\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]
1	3	5,02	1,67		
2	3	5,04	1,68		
3	3	5,04	1,68		
4	3	5,00	1,67		
5	3	5,01	1,67		
6	3	5,02	1,67	1,67	$1{,}80\cdot10^{-5}$
7	3	5,02	1,67		
8	3	4,98	1,66		
9	3	5,00	1,67		
10	3	5,00	1,67		

Start and stop at the zero crossing

Accuracy of reading the stop watch: 1/100 s

26.08.2026
N. Sycheva

Measuring the spring constant

m [g]	Nr.	number of oscillations n	measured time t [s]	period of oscillation T [s]	mean value $\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]
50	1	3	2,70	0,90		
	2	3	2,84	0,95	0,907	$2{,}3\cdot10^{-2}$
	3	3	2,62	0,87		
100	1	3	3,76	1,25		
	2	3	3,67	1,22	1,233	$9\cdot10^{-3}$
	3	3	3,70	1,23		
150	1	3	4,48	1,49		
	2	3	4,50	1,50	1,498	$3\cdot10^{-3}$
	3	3	4,50	1,50		
200	1	3	5,04	1,68		
	2	3	5,02	1,67	1,674	$3\cdot10^{-3}$
	3	3	5,01	1,67		
250	1	3	5,53	1,84		
	2	3	5,46	1,82	1,849	$1{,}8\cdot10^{-2}$
	3	3	5,65	1,88		

Measuring the period of oscillation as a function of the mass.
Time measurement started and stopped at zero passing

Measurement of the gravitational acceleration

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Δm [g]	deflection x [mm]	error of reading Δx [mm]
0	177	1
50	342	1
100	505	1
150	669	1
200	850	1
250	985	1

Measurement of deflection as a function of mass

3 Data analysis and results

3.1 Error propagation and statistical basics

For the analysis the followin formulas have been used:

- **Mean value of the osciallation period:**

$$\bar{T} = \frac{1}{n}\sum_{i=1}^{n} T_i \tag{7}$$

- **Mean error of the mean value (standard deviation):**

$$\sigma_{\bar{T}} = \sqrt{\frac{1}{n(n-1)}\sum_{i=1}^{n}(T_i - \bar{T})^2} \tag{8}$$

- **Error propagation for the squared oscillation period T^2:** As the graphical analysis of the squared osciálltion period T^2 is used we have to use gaussian error propagation to determine the uncertainty σ_{T^2} of T^2:

$$\sigma_{T^2} = \left|\frac{\mathrm{d}(T^2)}{\mathrm{d}T}\right| \cdot \sigma_{\bar{T}} = 2 \cdot \bar{T} \cdot \sigma_{\bar{T}} \tag{9}$$

3.2 Exercise 1: Comparison of the error for the different measuring methods

From the 10 measurement series with each measurement being $n = 3$ osciallations for a mass of $m = 200\,\mathrm{g}$ result in the following numbers:

- **Start/stop at maximum displacement:**

$$\begin{aligned} \bar{T}_{\text{max}} &= 1{,}587\,\mathrm{s} \\ \sigma_T &= 2{,}3 \cdot 10^{-2}\,\mathrm{s} \quad \text{(Error of individual measurement)} \\ \sigma_{\bar{T}_{\text{max}}} &= 7{,}7 \cdot 10^{-3}\,\mathrm{s} \quad \text{(Error of the mean value)} \end{aligned}$$

- **Start/stop at zero passing:**

$$\begin{aligned} \bar{T}_{\text{null}} &= 1{,}670\,\mathrm{s} \\ \sigma_T &= 6{,}0 \cdot 10^{-3}\,\mathrm{s} \quad \text{(Error of individual measurement)} \\ \sigma_{\bar{T}_{\text{null}}} &= 1{,}8 \cdot 10^{-3}\,\mathrm{s} \quad \text{(Error of the mean value)} \end{aligned}$$

Result: The relativ error of the mean value with measuring at zero passing is $\frac{\sigma_{\bar{T}}}{\bar{T}} \approx 0{,}11\%$, while measuring at maximum displacement yields $\approx 0{,}49\%$. The measurement at zero passing is therefore more accurate by a factor of 4.

3.3 Exercise 2: Graphically determining the spring constant D

In determining the spring constant we have plotted the squared oscillation amplitude T^2 versus the mass m. With the measurements and equation (9) we can determine the error bars with σ_{T^2}. The results are summarized in table 1. From graphical linear regression with $y = a \cdot m + b$ for $y = T^2$ we determine the slope a:

$$a = \frac{\Delta T^2}{\Delta m} \approx 13.1\,\frac{\mathrm{s}^2}{\mathrm{kg}} \tag{10}$$

m [kg]	$\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]	T^2 [s^2]	$\sigma_{T^2} = 2\bar{T}\sigma_{\bar{T}}$ [s^2]
0,050	0,907	$2{,}3 \cdot 10^{-2}$	0,823	0,042
0,100	1,233	$9{,}0 \cdot 10^{-3}$	1,520	0,022
0,150	1,498	$3{,}0 \cdot 10^{-3}$	2,244	0,009
0,200	1,674	$3{,}0 \cdot 10^{-3}$	2,802	0,010
0,250	1,849	$1{,}8 \cdot 10^{-2}$	3,419	0,067

Table 1: Values for osciälltion period T and calculated T^2 with error propagation.

Using best fit and worst fit line we determine the uncertainty of the slope to be $\sigma_a \approx 0.5\,\frac{\mathrm{s}^2}{\mathrm{kg}}$. From equation (4) we determine $a = \frac{4\pi^2}{D}$. From that we determine the spring constant D to be:

$$D = \frac{4\pi^2}{a} = \frac{4\pi^2}{13.1\,\frac{\mathrm{s}^2}{\mathrm{kg}}} \approx 3.01\,\frac{\mathrm{N}}{\mathrm{m}} \tag{11}$$

The error of the spring constant σ_D is calculated via error propagation:

$$\sigma_D = \left|\frac{\mathrm{d}D}{\mathrm{d}a}\right| \cdot \sigma_a = \frac{4\pi^2}{a^2} \cdot \sigma_a = \frac{4\pi^2}{13.1\,\frac{\mathrm{s}^2}{\mathrm{kg}}} \cdot 0.5\,\frac{\mathrm{s}^2}{\mathrm{kg}} \approx 0.11\,\frac{\mathrm{N}}{\mathrm{m}} \tag{12}$$

Endresult: Spring constant

The spring used in the experiment has a spring constant of:

$$D = (3.01 \pm 0.11)\,\frac{\mathrm{N}}{\mathrm{m}} \tag{13}$$

3.4 Exercise 3: Graphically determining the earths acceleration g

The displacement x in relation to the mass m can be viewed as a linear function $x(m)$ and we now have to determine its slope.

m [kg]	Displacement x [m]	Error Δx [m]
0,050	0,177	0,001
0,100	0,342	0,001
0,150	0,505	0,001
0,200	0,669	0,001
0,250	0,850	0,001

Table 2: Measurements of displacement x as a function of mass m.

From best fit line $x = a' \cdot m + x_0$ we can determine a' to he:

$$a' = \frac{\Delta x}{\Delta m} \approx 3{,}36\,\frac{\mathrm{m}}{\mathrm{kg}} \quad \text{mit einem abgeschätzten Steigungsfehler } \sigma_{a'} \approx 0{,}04\,\frac{\mathrm{m}}{\mathrm{kg}} \tag{14}$$

From hooks law ($x = \frac{g}{D} \cdot m$) we determine the slope to be $a' = \frac{g}{D}$. From that we get the gravitational accelaration g to be:

$$g = D \cdot a' = 3.01\,\frac{\mathrm{N}}{\mathrm{m}} \cdot 3.36\,\frac{\mathrm{m}}{\mathrm{kg}} \approx 10.13\,\frac{\mathrm{m}}{\mathrm{s}^2} \tag{15}$$

As D as well as a' have an error we need to calculate the the resulting error σ_g via gaussian error propagation for two variables:

$$\sigma_g = \sqrt{\left(\frac{\partial g}{\partial D} \cdot \sigma_D\right)^2 + \left(\frac{\partial g}{\partial a'} \cdot \sigma_{a'}\right)^2} \tag{16}$$

$$= \sqrt{(a' \cdot \sigma_D)^2 + (D \cdot \sigma_{a'})^2} \tag{17}$$

$$= \sqrt{(3{,}365 \cdot 0{,}06)^2 + (3{,}041 \cdot 0{,}04)^2} \approx 0{,}24 \,\frac{\mathrm{m}}{\mathrm{s}^2} \tag{18}$$

Endresult: Gravitational acceleration

$$g = (10{,}2 \pm 0{,}2)\,\frac{\mathrm{m}}{\mathrm{s}^2} \tag{19}$$

4 Conclusion and discussion

4.1 Comparison with real values

Compared to the real value for g we are slightly off